Housing Price Forecasting

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1/17/19

Framework & Data Sources

Goals

Identify Top 5 Zipcodes for Real Estate Investment in the Bay Area

- *Forecast expected growth over a relevant time period*
- *Control for risk*

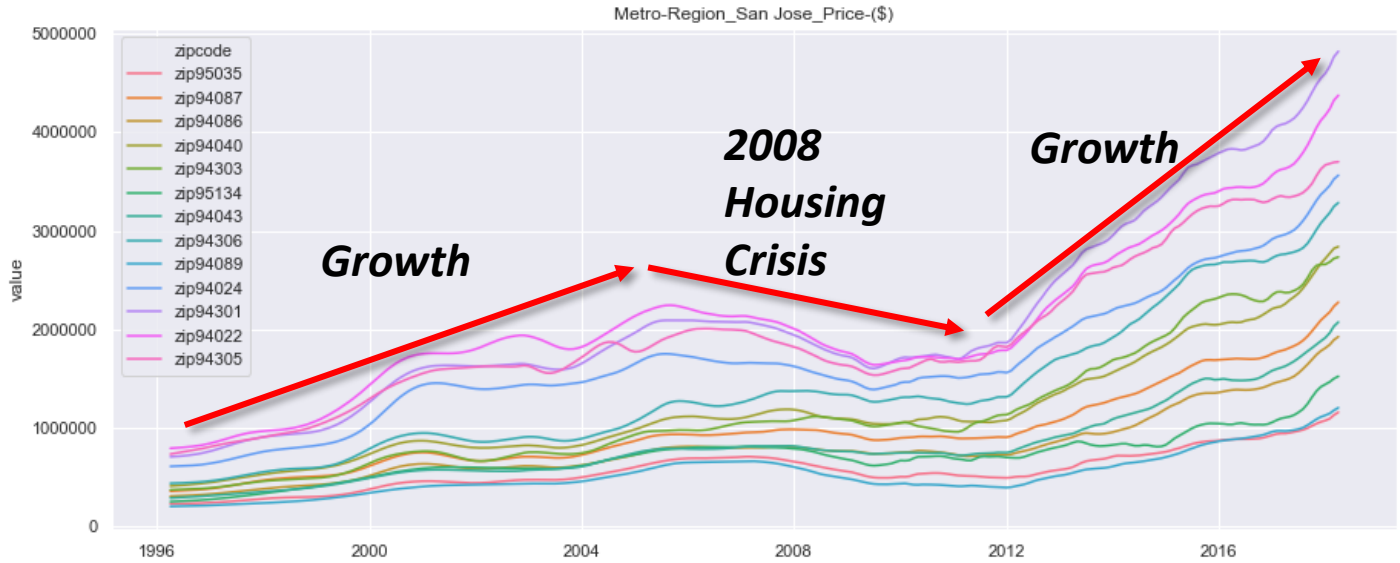
Methods

ARIMA forecasting *- auto-regressive and moving-average based statistical analysis*

Data Sources

- **Zillow Price Index Data: 22 years of data (1996-2018)**
  - Monthly, already seasonally adjusted
  - Zip code level
- **Zillow's Buyer-Seller Index (BSI)**

Bay Area EDA and Home Price Trends



Initial Observations

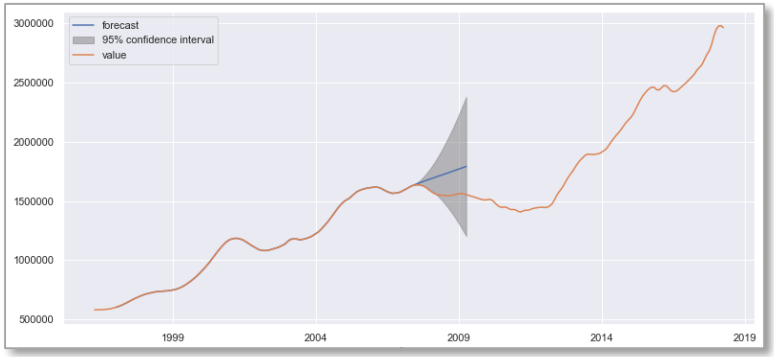
- **Three phases of housing price changes over the last 22 years**
  - Growth: 1996-2005
  - Flat to Decline: 2005-2010
  - Growth 2010-Present
- **Growth Heterogeneity:** Each zipcode experienced these phases very differently – some saw no price declines during the crisis, some saw significant ones. The last 8 years of price growth has impacted some regions much more intensely than others

ARIMA Projections - Preamble

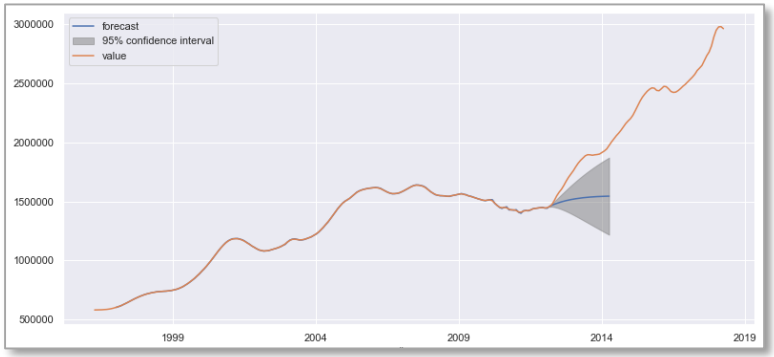
Considerations

- **ARIMA projections can be thwarted by sudden bubbles or price bursts**
  - **Bubble Bursts**: actual price stays within confidence interval, but underperforms
  - **Bubble Starts**: prices jump so quickly it exceeds the 95% confidence interval projections
- **ARIMA projections are highly effective at projecting onto relatively stable price growth trends**

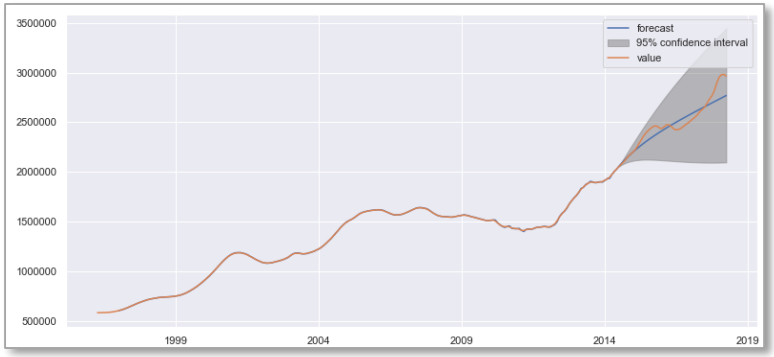
Bubble Bursts:



Bubble Starts:



Stable Prices:



Note: 94010 Zipcode Data Shown

ARIMA Projections - Results

We have identified 6 zip codes in the San Francisco Bay Area (of 153) for highest-return investment

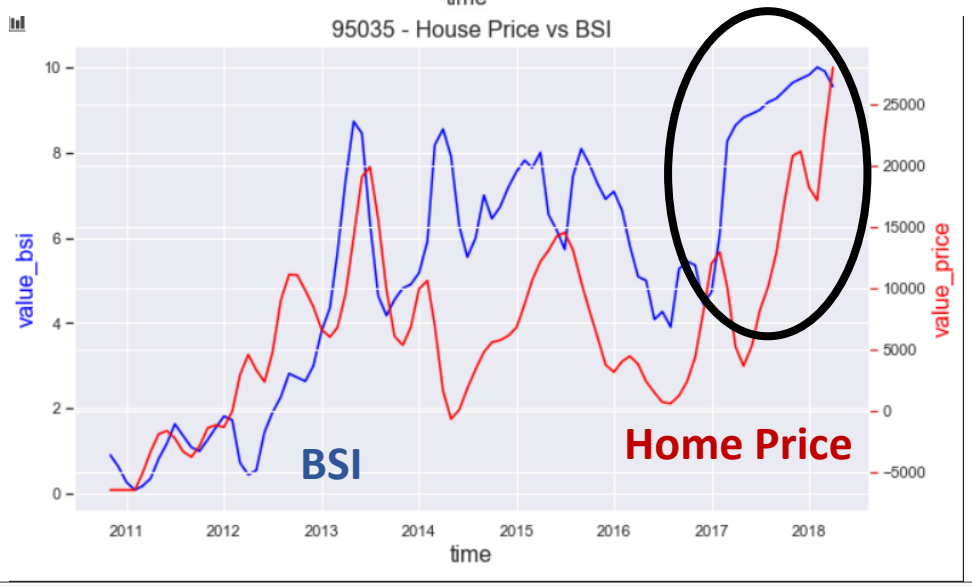
- **Option**: Deprioritize 94572 if the risk profile is too volatile
- **Note**: These zip codes are significantly smaller than the median zip code

Zipcode	Forecast 2-Yr Growth	97.5% Lower-Bound
94572	**47 %**	-4 %
94043	**39 %**	17 %
95035	**38 %**	8 %
94089	**31 %**	5 %
94063	**30 %**	1 %
94087	**28 %**	6 %

ARIMA Projections – Supporting Data

Buyer-Seller Index

- **BSI is at historically high levels, supporting the thesis of our high-growth forecasts**
- **Slight dips in BSI over the past 2 months of data – keep an eye on it**
- **BSI Index –** measuring the ration of buyers to sellers in a market (high = more buyers)
  - Not available for all zip codes
  - Showing two of the top 10 high-growth zips



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Client Recommendation Summary

Recommendations

1. **Prioritized list of zip codes with 2-year expected price growth and lower-bound growth estimates**
   - If risk-averse, choose top 5 zip codes shown
   - If risk-averse, choose zip codes 2-6 shown as the top region is price-volatile

2. **Beware bubbles!!**
   - ARIMA forecast method cannot predict bubbles forming or bubbles bursting
   - Sudden changes in housing prices due to exogenous factors (e.g. recession, home price crisis) may exceed our calculated lower-bound growth rate

3. **Monitor Buyer-Seller Index – currently at historical high levels, but has dipped in recent months**
   - Continued slide in the BSI could be an indicator of slipping demand

Next Analytic Steps

1. **Test sensitivity of ARIMA projections to last few months of data**
   - Last-period trends seem to have an outside impact on forecast pricing values
2. **Heatmap of zip codes and price growth projections**
3. **Evaluate correlated predictors (e.g. BSI type indicators)**

Discussion & Questions

Thanks!